

SQL Execution Plan -

Hi friends today we will learn how sql make execution plan visually explained them.



It is an exactly pin point tool to find why your query is slow

Large query have lot of joins and aggregation so it is slow
So to improve performance we used indexes

Where exactly we are going build the indexes in which table in which columns
So for this we have to understand where exactly is the problem is it in sorting the data or joining tables or in aggregating data or by sorting data.

Now in order to answer there questions we have execution plan

How the data base exactly process your query step by step
It shows where is exactly performance issue.

So, execution plan is like a window to understand how the sql db exactly thinks.
And once we understand then we can make a right decision fro building an index to improve performance or solving issues.

```

JOIN DimSalesChannel pc ON s.SalesChannelKey = pc.SalesChannelKey
GROUP BY pc.SalesChannel, YEAR(s.OrderDate), DATEPART(QUARTER, s.OrderDate);

-- Query 6: Customer demographics and purchase behavior by region
SELECT c.CustomerKey, c.Gender, c.MaritalStatus, st.SalesTerritoryRegion,
       COUNT(s.SalesOrderNumber) AS OrderCount, SUM(s.SalesAmount) AS TotalSpent
FROM FactInternetSales s
JOIN DimCustomer c ON s.CustomerKey = c.CustomerKey
JOIN DimSalesTerritory st ON s.SalesTerritoryKey = st.SalesTerritoryKey
GROUP BY c.CustomerKey, c.Gender, c.MaritalStatus, st.SalesTerritoryRegion;

-- Query 7: Revenue by product category and year for the last decade
SELECT pc.ProductCategoryKey, YEAR(s.OrderDate) AS SalesYear,
       SUM(s.SalesAmount) AS AnnualRevenue
FROM FactInternetSales s
JOIN DimProductCategory pc ON s.ProductCategoryKey = pc.ProductCategoryKey
WHERE s.OrderDateKey BETWEEN 20100101 AND 20201231
GROUP BY pc.ProductCategoryKey, YEAR(s.OrderDate);

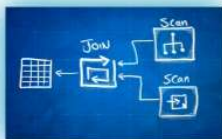
-- Query 8: Monthly trend analysis by customer segment and reseller
SELECT c.CustomerSegment, r.ResellerKey, MONTH(s.OrderDate) AS SalesMonth,
       SUM(s.SalesAmount) AS MonthlyRevenue
FROM FactResellerSales s
JOIN DimCustomer c ON s.CustomerKey = c.CustomerKey
JOIN DimReseller r ON s.ResellerKey = r.ResellerKey
GROUP BY c.CustomerSegment, r.ResellerKey, MONTH(s.OrderDate);

-- Query 9: Product sales performance in each territory
SELECT st.SalesTerritoryKey, st.SalesTerritoryCountry, ps.ProductSubcategoryKey,
       COUNT(s.SalesOrderNumber) AS SalesCount, SUM(s.SalesAmount) AS SalesTotal
FROM FactResellerSales s
JOIN DimSalesTerritory st ON s.SalesTerritoryKey = st.SalesTerritoryKey
JOIN DimProductSubcategory ps ON s.ProductSubcategoryKey = ps.ProductSubcategoryKey
GROUP BY st.SalesTerritoryKey, st.SalesTerritoryCountry, ps.ProductSubcategoryKey;

-- Query 10: Quarterly sales trends for high-value customers
SELECT c.CustomerKey, DATEPART(QUARTER, s.OrderDate) AS SalesQuarter,
       SUM(s.SalesAmount) AS QuarterlySales
FROM FactInternetSales s
JOIN DimCustomer c ON s.CustomerKey = c.CustomerKey
WHERE s.SalesAmount > 1000
GROUP BY c.CustomerKey, DATEPART(QUARTER, s.OrderDate);

-- Query 11: Employee sales performance by quarter and year

```

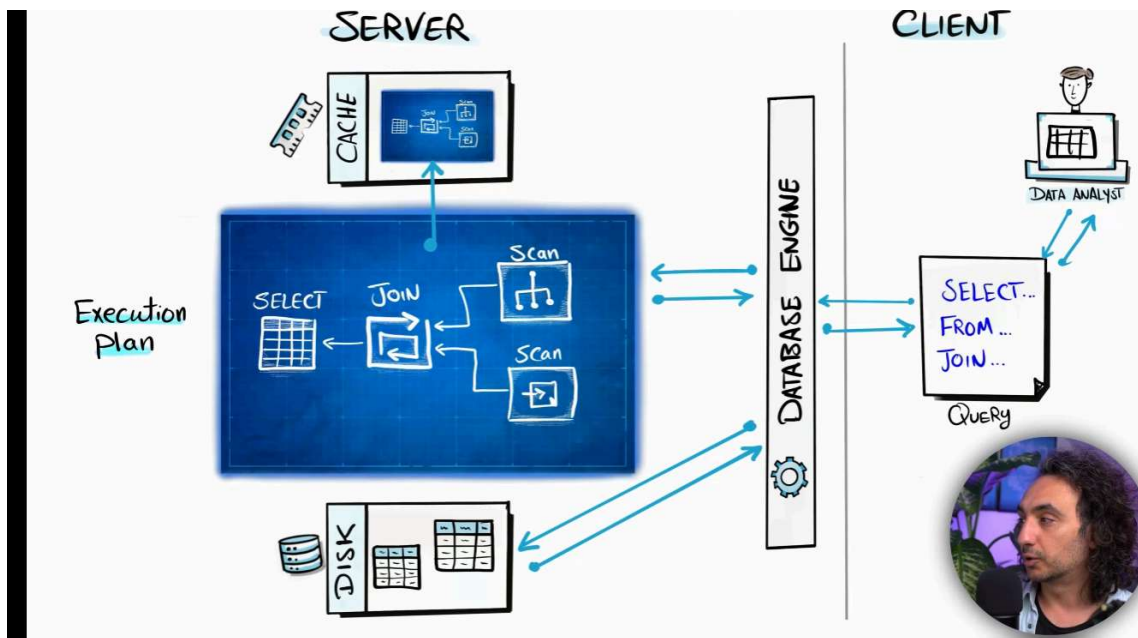


EXECUTION PLAN

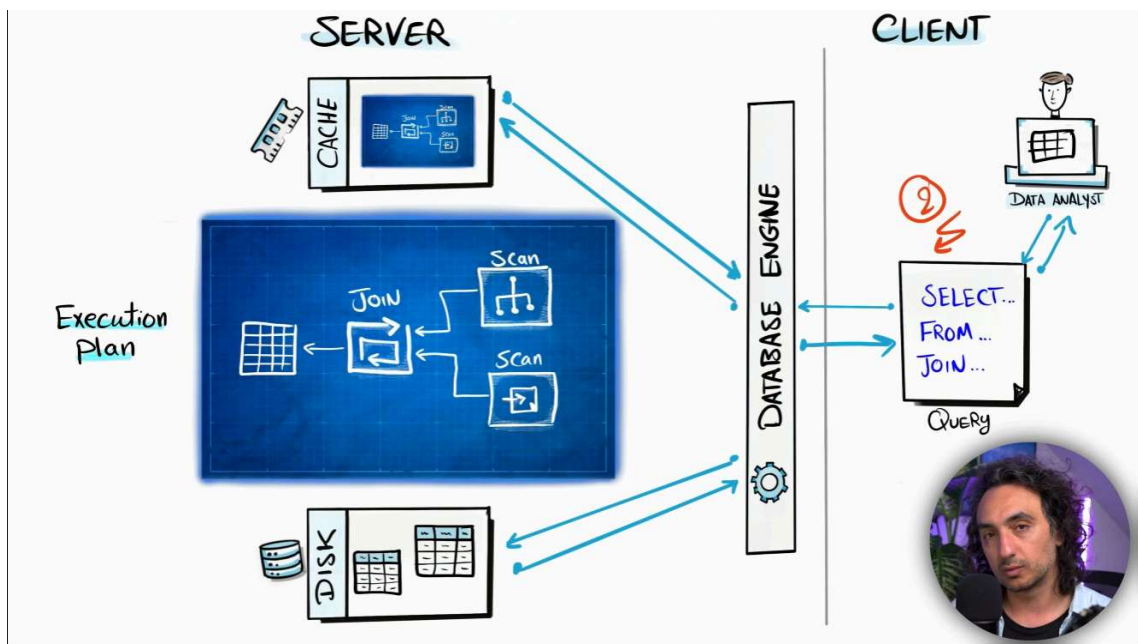
Roadmap generated by a database
on how it will execute your query step by step

Execution plan like finding roadmap over map to reach at destination.
It check how to scan table like how to get data from tables like to get multiple tables.

To decide it's full table scan or index scan then decide which type of join loop
join single join then select statement.

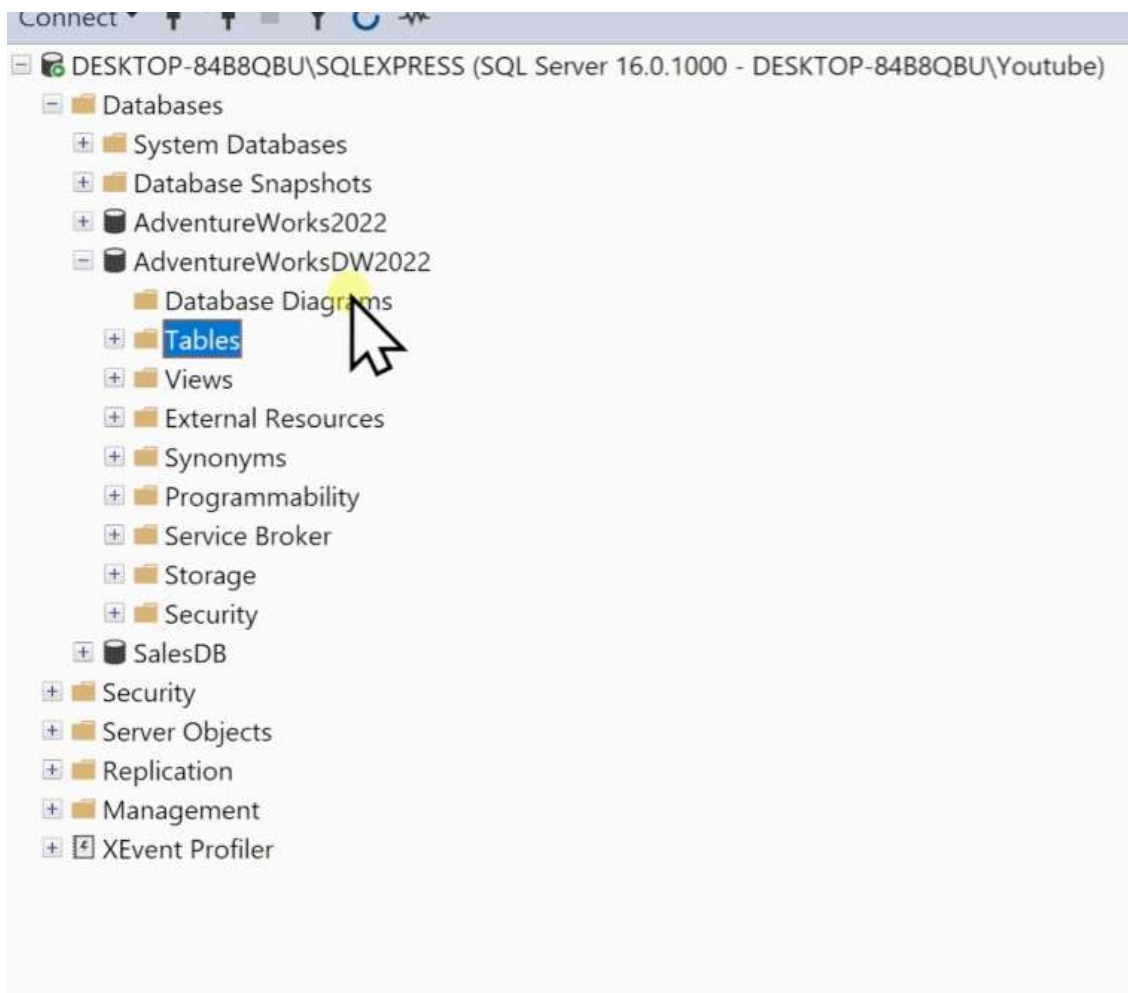


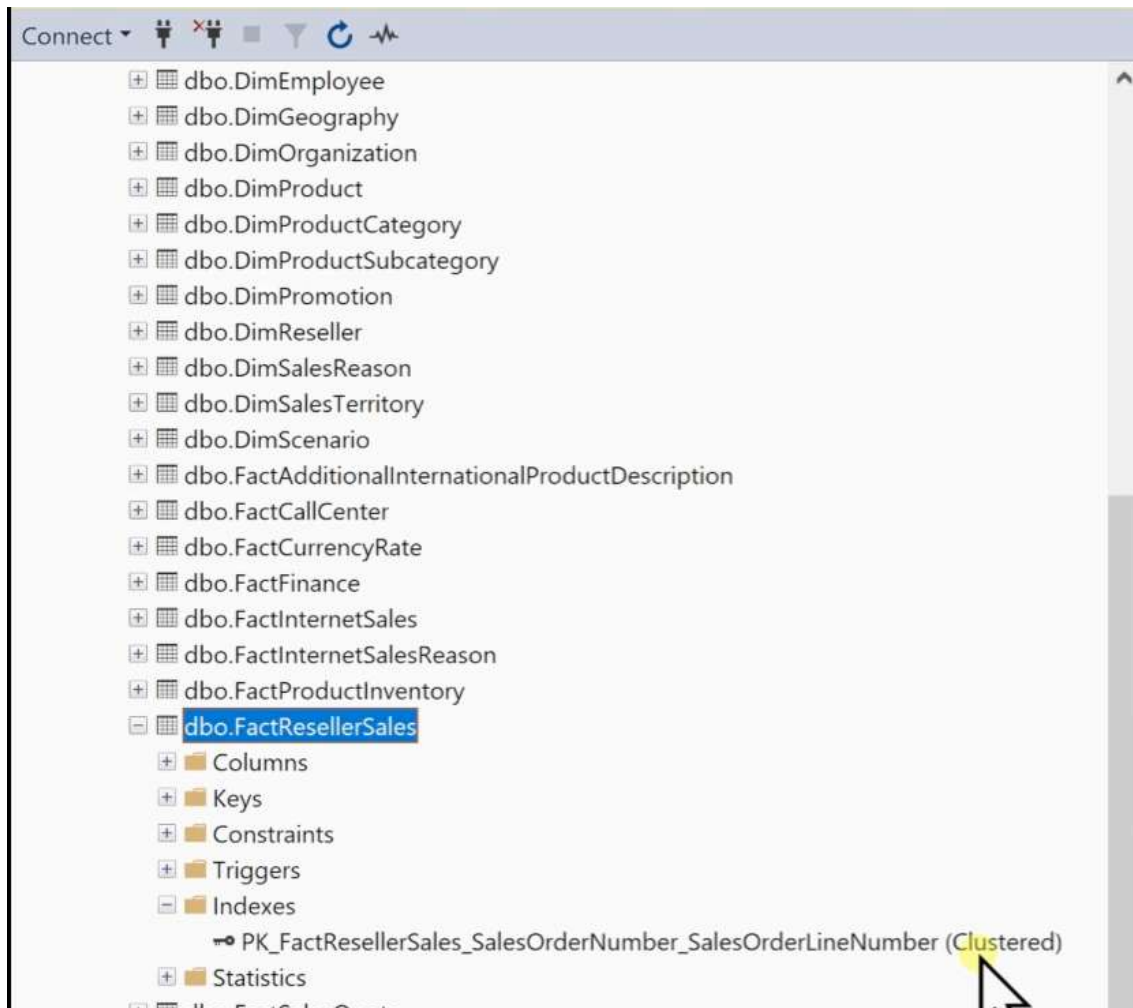
When you run same query again - takes help from cache store execution plan



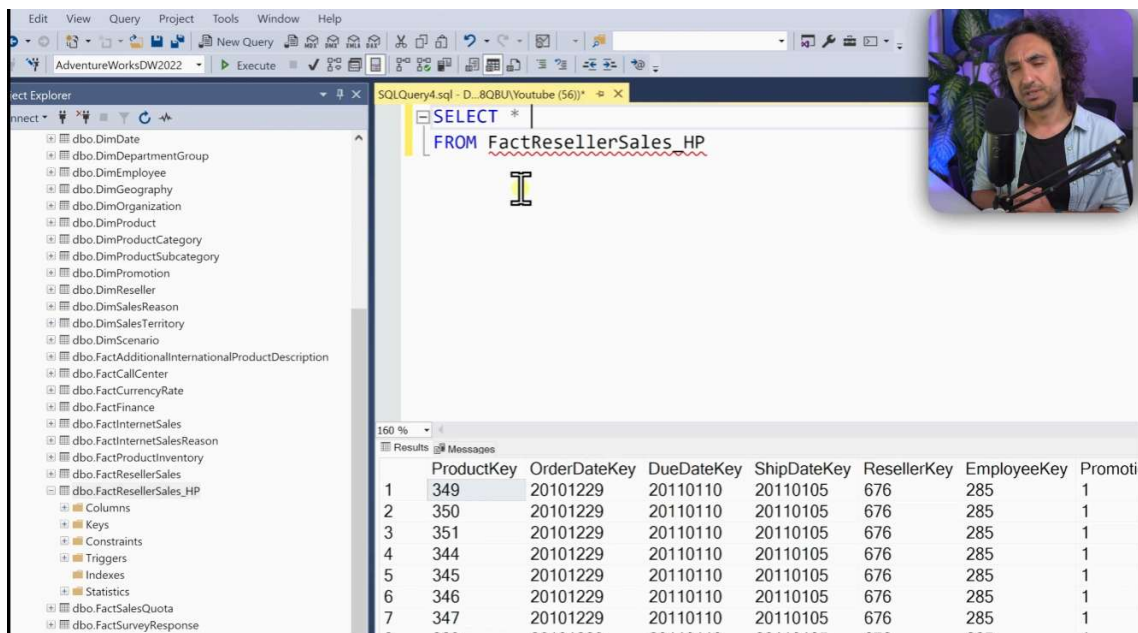
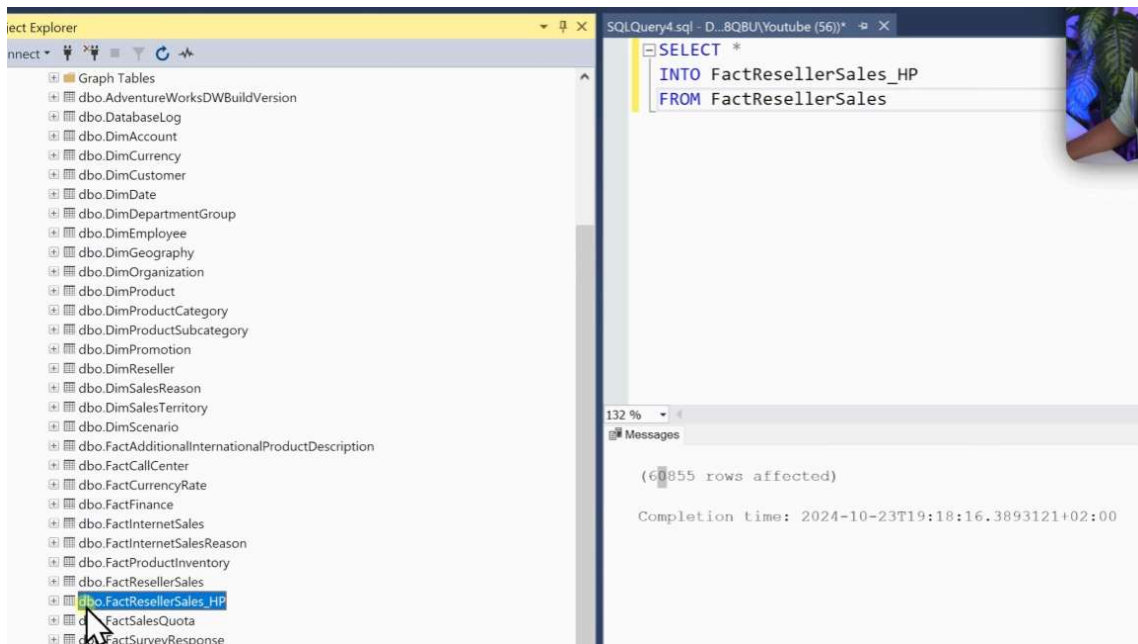
Execution Plan

Basics





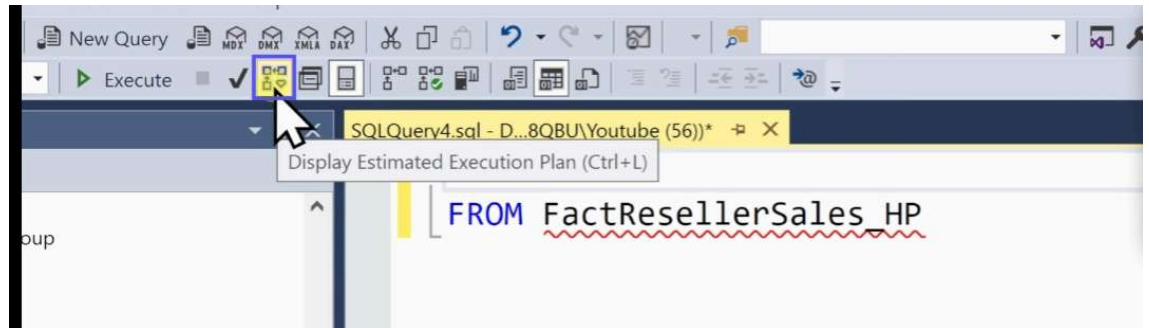
Create same mirror table I mean creating same table without index



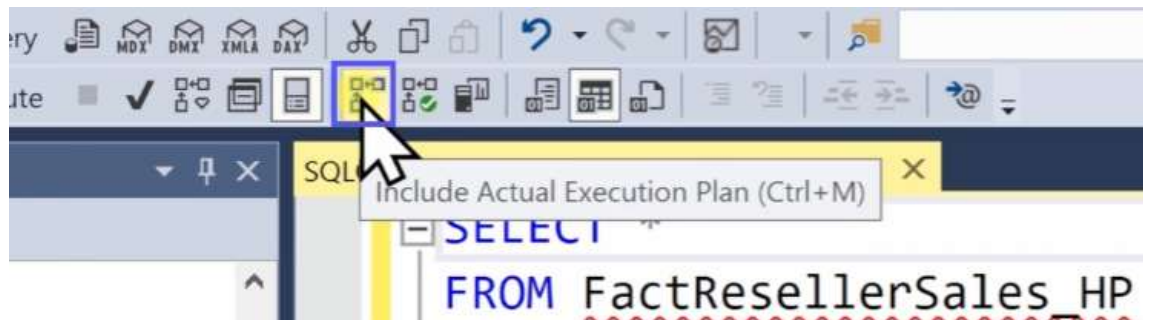
How to check execution plan of this query.

Go to at top tool bar

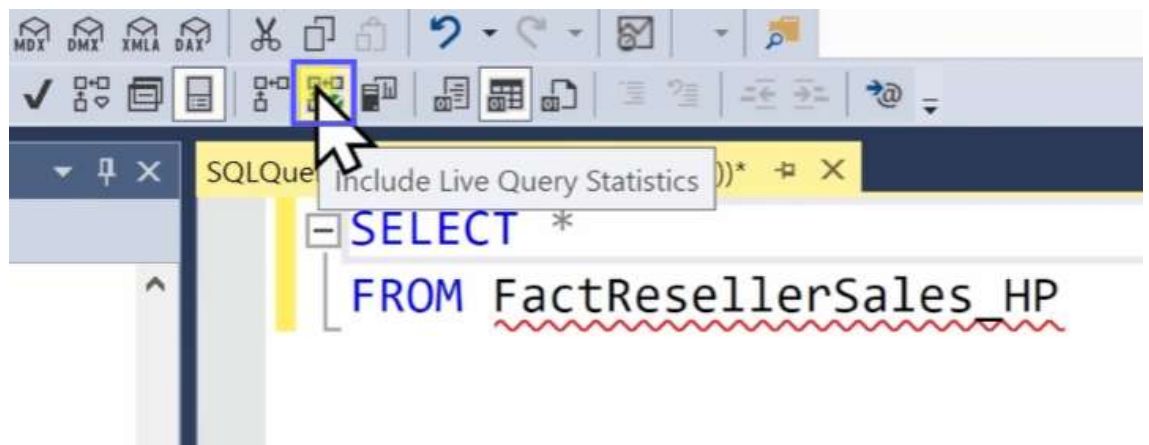
1.



2.



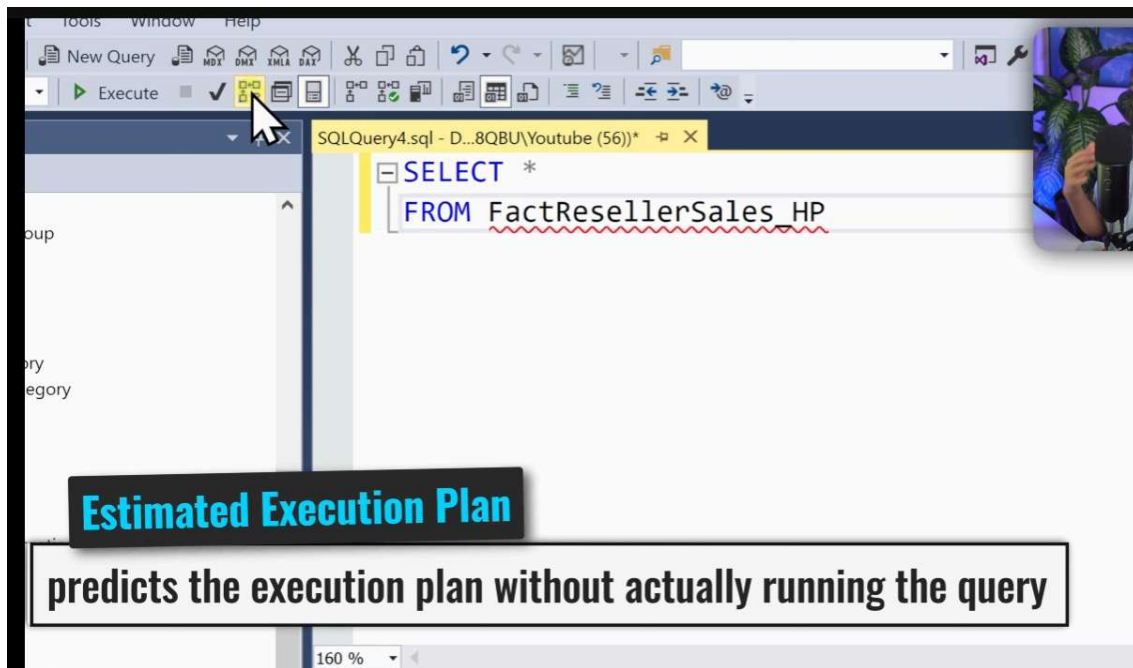
3.



Now check what are difference between them let's go with first one.

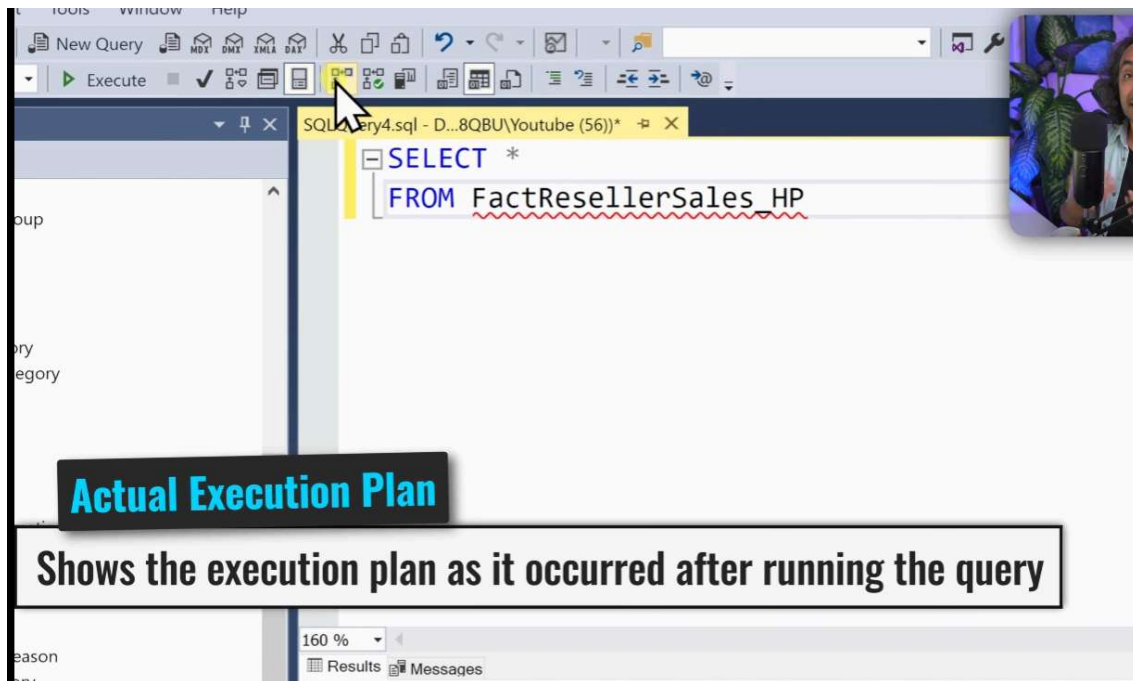
First one is guess only without execution the query

It is before execution of your query.



Second one is actual one show execution plan that show you that is used to process your query

It is after execution your query



Third one is live execution plan - it is while execution of your query.

ry
egory

Live Execution Plan

Shows the real-time execution flow as the query runs.

160 %
5:54 / 32:15
Executing
Results
Messages

First one

Tools window Help

New Query
Execute
✓
MDX
DMX
XMLA
DAY

SQLQuery4.sql - D...8QBU\Youtube (56))*

SELECT *
FROM FactResellerSales_HP

160 %
Messages
Execution plan

Query 1: Query cost (relative to the batch
SELECT * FROM FactResellerSales_HP

SELECT
Cost: 0 %

Table Scan
[FactReseller...
Cost: 100 %

Second one

The screenshot shows the SQL Server Enterprise Manager interface. The query editor displays the following SQL statement:

```
SELECT *  
FROM FactResellerSales_HP
```

The query is executed, and the execution plan is shown. The plan consists of a single operator: a Table Scan for the table [FactResellerSales_HP]. The cost of the query is 100%, and the execution time is 0.053s. The plan also shows that 60855 rows were scanned, representing 100% of the table.

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM FactResellerSales_HP

Table Scan
[FactResellerSales_HP]
Cost: 100 %
0.053s
60855 of 60855 (100%)

Third one

The screenshot shows the SQL Server Enterprise Manager interface. The query editor displays the following SQL statement:

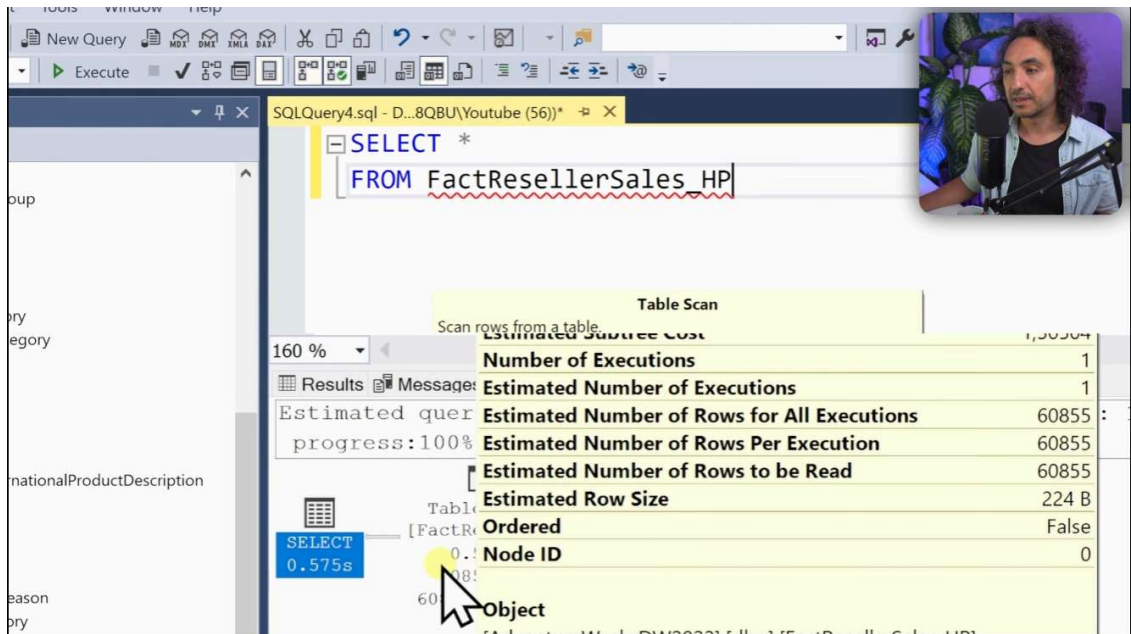
```
SELECT *  
FROM FactResellerSales_HP
```

The query is executed, and the execution plan is shown. The plan consists of a single operator: a Table Scan for the table [FactResellerSales_HP]. The cost of the query is 100%, and the execution time is 0.575s. The plan also shows that 60855 rows were scanned, representing 100% of the table.

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM FactResellerSales_HP

Table Scan
[FactResellerSales_HP]
Cost: 100 %
0.575s
60855 of 60855 (100%)

A video inset in the top right corner shows a man with curly hair, wearing a light blue shirt, speaking into a microphone.



SQLQuery4.sql - D:\8QBU\Youtube (56)*

```
SELECT *
FROM FactResellerSales_HP
```

Table Scan

Scan rows from a table.

Estimated Statistics	
Estimated Number of Executions	1
Estimated Number of Rows for All Executions	60855
Estimated Number of Rows Per Execution	60855
Estimated Number of Rows to be Read	60855
Estimated Row Size	224 B
Ordered	False
Node ID	0
Object	

Estimated query progress: 100%

Table Scan [FactResellerSales_HP] 0.575s

Estimated vs Actual Plans

If the predictions don't match the Actual Execution Plan, this indicates issues like inaccurate statistics or outdated indexes, leading to poor performance.

Now check execution plan for clustered index and heap.

Actual execution plan

Heap execution plan - reads from right to left

Query 1: Query

```
SELECT * FROM FactResellerSales_HP
```

Table Scan

Scan rows from a table.

Physical Operation: Table Scan

Logical Operation: Table Scan

Actual Execution Mode: Row

Estimated Execution Mode: Row

Storage: RowStore

Actual Number of Rows Read: 60855

Actual Number of Rows for All Executions: 60855

Actual Number of Batches: 0

Estimated I/O Cost: 1,23802

Estimated Operator Cost: 1,30504 (100%)

Estimated CPU Cost: 0,067019

Estimated Subtree Cost: 1,30504

Number of Executions: 1

Estimated Number of Executions: 1

Estimated Number of Rows for All Executions: 60855

Estimated Number of Rows Per Execution: 60855

Estimated Number of Rows to be Read: 60855

Estimated Row Size: 224 B

Actual Rebinds: 0

Actual Rewinds: 0

Ordered: False

Node ID: 0

Object: [AdventureWorksDW2022].[dbo].[FactResellerSales_HP]

Output List

- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[ProductKey]
- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[OrderDateKey]
- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[ShipDateKey]
- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[ResellerKey]
- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[EmployeeKey]
- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[PromotionKey]
- [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[SalesTerritoryKey]

Check by right click on table and got to properties read would be easy.

Query 1: Query cost (rel)

```
SELECT * FROM FactResellerSales_HP
```

Table Scan

Cost: 100...

0,053s

60855 of 60855 (100%)

Actual Number of Rows for All Executions: 60855

Actual Number of Rows Read: 60855

Actual Rebinds: 0

Actual Rewinds: 0

Actual Time Statistics

Defined Values

Description: Scan rows from a table.

Estimated CPU Cost: 0,067019

Estimated Execution Mode: Row

Estimated I/O Cost: 1,23802

Estimated Number of Executions: 1

Estimated Number of Rows for All Executions: 60855

Estimated Number of Rows Per Execution: 60855

Estimated Number of Rows to be Read: 60855

Estimated Operator Cost: 1,30504 (100%)

Estimated Rebinds: 0

Estimated Rewinds: 0

Estimated Row Size: 224 B

Estimated Subtree Cost: 1,30504

Forced Index: False

ForceScan: False

Logical Operation: Table Scan

Node ID: 0

NoExpandHint: False

Number of Executions: 1

Object: [AdventureWorksDW2022].[dbo].[FactResellerSales_HP]

Ordered: False

Output List: [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[ProductKey], [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[OrderDateKey], [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[ShipDateKey], [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[ResellerKey], [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[EmployeeKey], [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[PromotionKey], [AdventureWorksDW2022].[dbo].[FactResellerSales_HP].[SalesTerritoryKey]

Physical Operation: Table Scan

Storage: RowStore

Logical Operation: Relational algebraic operator this node represents. For rows of type PLAN_ROWS only.

Clustered index execution plan

SQLQuery2.sql - D:\8QBU\Youtube (52)*

SQLQuery1.sql - D:\8QBU\Youtube (58)*

SELECT *

FROM FactResellerSales

Clustered Index Scan (Clustered)

Scanning a clustered index, entirely or only a range.

Physical Operation	Clustered Index Scan
Logical Operation	Clustered Index Scan
Actual Execution Mode	Row
Estimated Execution Mode	Row
Storage	RowStore
Actual Number of Rows Read	60855
Actual Number of Rows for All Executions	60855
Estimated I/O Cost	1,241
Estimated Operator Cost	1,30874 (100%)
Estimated CPU Cost	0,0670975
Estimated Subtree Cost	1,308
Number of Executions	1
Estimated Number of Executions	60855
Estimated Number of Rows for All Executions	60855
Estimated Number of Rows Per Execution	60855
Estimated Number of Rows to be Read	60855
Estimated Row Size	224 B
Actual Rebinds	0
Actual Rewinds	0
Ordered	False
Node ID	0
Object	[AdventureWorksDW2022].[dbo].[FactResellerSales]
Output List	[AdventureWorksDW2022].[dbo].[FactResellerSales].ProductKey, [AdventureWorksDW2022].[dbo].[FactResellerSales].OrderDateKey, [AdventureWorksDW2022].[dbo].[FactResellerSales].ProductID

Results

Messages

Execution plan

Query 1: Query

SELECT * FROM FactResellerSales

Cost: 0 %

Clustered Index Scan (Clustered)

Cost: 0 %

60855

60855

Properties

Current connection parameters

Aggregate Status

Connection failures

Elapsed time

Finish time

Name

Rows returned

Start time

State

Connection

Connection name

Connection Details

Connection elapsed time

Connection encryption

Connection finish time

Connection rows returned

Connection start time

Connection state

Display name

Login name

Server name

Server version

Session Tracing ID

SPID

TDS protocol version

Name

The name of the connection.



Index Scan

Scans all data in an index to find matching rows.

SQLQuery2.sql - D:\8QBU\Youtube (52)*

SQLQuery1.sql - D:\8QBU\Youtube (58)*

SELECT *

FROM FactResellerSales

Clustered Index Scan (Clustered)

Scanning a clustered index, entirely or only a range.

Physical Operation	Clustered Index Scan
Logical Operation	Clustered Index Scan
Actual Execution Mode	Row
Estimated Execution Mode	Row
Storage	RowStore
Actual Number of Rows Read	60855
Actual Number of Rows for All Executions	60855
Estimated I/O Cost	1,241
Estimated Operator Cost	1,30874 (100%)
Estimated CPU Cost	0,0670975
Estimated Subtree Cost	1,308
Number of Executions	1
Estimated Number of Executions	60855
Estimated Number of Rows for All Executions	60855
Estimated Number of Rows Per Execution	60855
Estimated Number of Rows to be Read	60855
Estimated Row Size	224 B
Actual Rebinds	0
Actual Rewinds	0
Ordered	False
Node ID	0
Object	[AdventureWorksDW2022].[dbo].[FactResellerSales]
Output List	[AdventureWorksDW2022].[dbo].[FactResellerSales].ProductKey, [AdventureWorksDW2022].[dbo].[FactResellerSales].OrderDateKey, [AdventureWorksDW2022].[dbo].[FactResellerSales].ProductID

Results

Messages

Execution plan

Query 1: Query cost (relative to the batch): 100

SELECT * FROM FactResellerSales

Cost: 0 %

Clustered Index Scan (Clustered)

Cost: 100 %

60855 of 60855 (100%)

Properties

Clustered Index Scan (Clustered)

Actual Execution Mode

Actual I/O Statistics

Actual Number of Batches

Actual Number of Rows for All Executions

Actual Number of Rows Read

Actual Rebinds

Actual Rewinds

Actual Time Statistics

Defined Values

Description

Estimated CPU Cost

Estimated Execution Mode

Estimated I/O Cost

Estimated Number of Executions

Estimated Number of Rows for All Executions

Estimated Number of Rows Per Execution

Estimated Number of Rows to be Read

Estimated Operator Cost

Estimated Rebinds

Estimated Rewinds

Estimated Row Size

Estimated Subtree Cost

Forced Index

ForceScan

Logical Operation

Node ID

NoExpandHint

Number of Executions

Object

Ordered

Output List

Parallel

Physical Operation

Storage

Actual Number of Rows Read

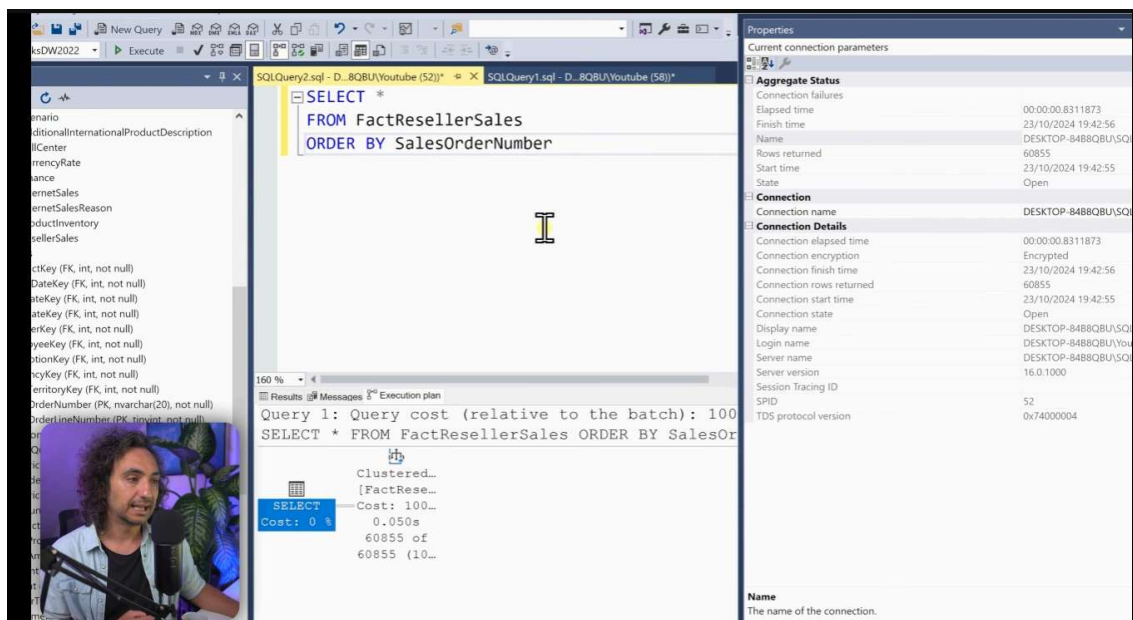
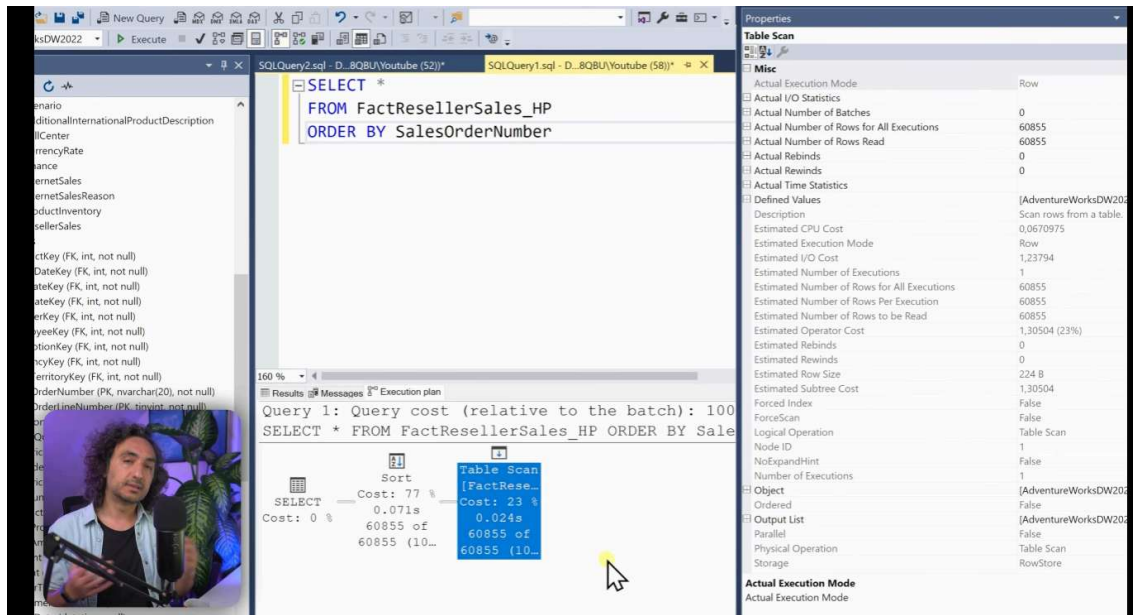
Number of rows read from a table or an index prior to applying a predicate filter. For

Execution Plan

Heap vs. Clustered Index

Execution Plan

Heap vs. Clustered Index



No extra step to do sort in clustered index because data is already sorted.

SQLQuery2.sql - D:\8QBU\Youtube (52)*

SQLQuery1.sql - D:\8QBU\Youtube (58)*

SELECT *

FROM FactResellerSales

ORDER BY SalesOrderNumber

Query 1: Query cost (relative to the batch): 100

SELECT * FROM FactResellerSales ORDER BY SalesOrderNumber

Clustered Index Scan (Clustered)

Cost: 0 %

0.048s

60855 of 60855 (100%)

Properties

Clustered Index Scan (Clustered)

Misc

Actual Execution Mode: Row

Actual I/O Statistics

Actual Number of Batches: 0

Actual Number of Rows for All Executions: 60855

Actual Number of Rows Read: 60855

Actual Rebinds: 0

Actual Rewinds: 0

Actual Time Statistics

Defined Values: [AdventureWorksDW2008]

Description: Scanning a clustered index

Estimated CPU Cost: 0.0670975

Estimated Execution Mode: Row

Estimated I/O Cost: 1.24164

Estimated Number of Executions: 1

Estimated Number of Rows for All Executions: 60855

Estimated Number of Rows Per Execution: 60855

Estimated Number of Rows to be Read: 60855

Estimated Operator Cost: 1.30874 (100%)

Estimated Rebinds: 0

Estimated Rewinds: 0

Estimated Row Size: 224 B

Estimated Subtree Cost: 1.30874

Forced Index: False

ForceScan: False

Logical Operation: Clustered Index Scan

Node ID: 0

NoExpandHint: False

Number of Rows: 60855

Object: [AdventureWorksDW2008].[FactResellerSales]

Ordered: False

Output List: [AdventureWorksDW2008].[FactResellerSales]

Parallel: False

Physical Operation: Clustered Index Scan

Storage: RowStore

Actual Execution Mode: Clustered Index Scan

Clustered Costs = 1,308

SQLQuery2.sql - D:\8QBU\Youtube (52)*

SQLQuery1.sql - D:\8QBU\Youtube (58)*

SELECT *

FROM FactResellerSales_HP

ORDER BY SalesOrderNumber

Query 1: Query cost (relative to the batch): 100

SELECT * FROM FactResellerSales_HP ORDER BY SalesOrderNumber

Sort

Cost: 77 %

0.071s

60855 of 60855 (100%)

Table Scan (FactResellerSales_HP)

Cost: 23 %

0.024s

60855 of 60855 (100%)

Properties

Table Scan

Misc

Actual Execution Mode: Row

Actual I/O Statistics

Actual Number of Batches: 0

Actual Number of Rows for All Executions: 60855

Actual Number of Rows Read: 60855

Actual Rebinds: 0

Actual Rewinds: 0

Actual Time Statistics

Defined Values: [AdventureWorksDW2008]

Description: Scan rows from a table

Estimated CPU Cost: 0.0670975

Estimated Execution Mode: Row

Estimated I/O Cost: 1.23794

Estimated Number of Executions: 1

Estimated Number of Rows for All Executions: 60855

Estimated Number of Rows Per Execution: 60855

Estimated Number of Rows to be Read: 60855

Estimated Operator Cost: 1.30504 (23%)

Estimated Rebinds: 0

Estimated Rewinds: 0

Estimated Row Size: 224 B

Estimated Subtree Cost: 1.30504

Forced Index: False

ForceScan: False

Logical Operation: Table Scan

Node ID: 0

NoExpandHint: False

Number of Rows: 60855

Object: [AdventureWorksDW2008].[FactResellerSales_HP]

Ordered: False

Output List: [AdventureWorksDW2008].[FactResellerSales_HP]

Parallel: False

Physical Operation: Table Scan

Storage: RowStore

Actual Execution Mode: Table Scan

Heap Costs = 1,308

Clustered Costs = 1,308

SQLQuery2.sql - D:\BQBU\Youtube (52)*

```
SELECT *
FROM FactResellerSales_HP
ORDER BY SalesOrderNumber
```

Properties

Sort

Misc

Actual Execution Mode: Row

Actual I/O Statistics

Actual Number of Batches: 0

Actual Number of Rows for All Executions: 60855

Actual Rebinds: 1

Actual Rewinds: 0

Actual Time Statistics

Description: Sort the input.

Distinct: False

Estimated CPU Cost: 4,43943

Estimated Execution Mode: Row

Estimated I/O Cost: 0,0112613

Estimated Number of Executions: 1

Estimated Number of Rows for All Executions: 60855

Estimated Number of Rows Per Execution: 60855

Estimated Operator Cost: 4,45069 (77%)

Estimated Rebinds: 0

Estimated Rewinds: 0

Estimated Row Size: 224 B

Estimated Subtree Cost: 5,75573

Logical Operation: Sort

Memory Fractions Input

Memory Usage

Estimated CPU Cost

Estimated CPU cost for this operator. For rows of type PLAN_ROWS only.

Heap Costs = 1,308 + 4,45 = 5,754

Clustered Costs = 1,308

High cost for sorting the data.

SQLQuery2.sql - D:\BQBU\Youtube (52)*

```
SELECT *
FROM FactResellerSales
ORDER BY SalesOrderNumber
```

Properties

Clustered Index Scan (Clustered)

Misc

Actual Execution Mode: Row

Actual I/O Statistics

Actual Number of Batches: 0

Actual Number of Rows for All Executions: 60855

Actual Number of Rows Read: 60855

Actual Rebinds: 0

Actual Rewinds: 0

Actual Time Statistics

Defined Values: (AdventureWorksDW2022)

Description: Scanning a clustered index.

Estimated CPU Cost: 0,0670975

Estimated Execution Mode: Row

Estimated I/O Cost: 1,24164

Estimated Number of Executions: 1

Estimated Number of Rows for All Executions: 60855

Estimated Number of Rows Per Execution: 60855

Estimated Rebinds: 1,30874 (100%)

Estimated Rewinds: 0

Estimated Row Size: 224 B

Estimated Subtree Cost: 1,30874

Forced Index: False

NoExpand

Number of Rows: 60855

Object: FactResellerSales

Ordered: True

Output List: (AdventureWorksDW2022)

Parallel: False

Physical Operation: Clustered Index Scan

Actual Execution Mode

Actual Execution Mode

Heap is slower than Clustered, because Database must perform extra work to sort rows

Heap Costs = 1,308 + 4,45 = 5,754

Clustered Costs = 1,308

The screenshot shows the SQL Server Enterprise Manager interface. On the left, a list of database objects is visible. The central pane displays a query: `SELECT * FROM FactResellerSales ORDER BY SalesOrderNumber`. Below the query, the execution plan is shown, indicating a 'Clustered Index Scan (Clustered)' operation. The right pane shows the properties for this operation, including 'Actual Execution Mode' (Row), 'Actual Number of Rows for All Executions' (60855), and 'Actual Number of Rows Read' (60855). The 'Index' property is highlighted, showing the index name: `[PK_FactResellerSales_SalesOrderNumber_SalesOrderNumber]`.

This screenshot is similar to the one above, showing the same query and execution plan. However, it includes a video inset in the bottom right corner showing a person speaking. The 'Index' property in the right pane is still highlighted, showing the index name: `[PK_FactResellerSales_SalesOrderNumber_SalesOrderNumber]`.

In execution plan you can find/check it which index is you are using in your query.

And this is important if you create new index and run query and check db is using your newly created index or not if not then this is wrong decisions about your index.

So each time your create your index and check through execution plan



Instead of pk we used any one of those column which we are using in our table.

Ex- carrier tracking number

SQLQuery2.sql - D:\8QBU\Youtube (52)*

SQLQuery1.sql - D:\8QBU\Youtube (58)*

```

SELECT *
FROM FactResellerSales_HP
WHERE CarrierTrackingNumber = '4911-403C-98'

```

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM [FactResellerSales_HP] WHERE [CarrierTrackingNumber] = '4911-403C-98'

Table Scan
[FactResellerSales_HP]
Cost: 100.00
0.007s
12 of 15 (80%)

Plans (Visually Explained) | SQL Hints | #SQL Course 40

SQL Backup Plan

We have table scan in execution plan

